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ARTICLE

Improving assessment of *Pandalus* stocks using a seasonal, size-structured assessment model with environmental variables. Part I: Model description and application

Jie Cao, Yong Chen, and R. Anne Richards

Abstract: *Pandalus* species display the following features that make it difficult to apply traditional age-based stock assessment models: (*i*) difficulty of determining age in the absence of hard parts retained through the molt; (*ii*) sex change in which individuals mature first as males and then transform to females; and (*iii*) potentially strong influence of environmental conditions on recruitment population dynamics. In this context, we propose a seasonal, size-structured assessment model dedicated to stock assessment of hermaphroditic Pandalidae. The modeling framework incorporates a submodel for changes of length at sex transformation and functions to incorporate environmental effects on recruitment dynamics. The model can be directly fitted to length-structured data, overcoming the length to age conversion problem. The model has a seasonal time step that allows it to account for seasonal variations in biological processes and fishing patterns. The model provides stock assessment outputs, such as fishing mortality and stock biomass estimates, and sex-specific abundance-at-length. The model is applied to the exploited shrimp stock of *Pandalus borealis* in the Gulf of Maine as an example of its utility. The model proposed in this study is flexible and generic and can be applied to many other exploited stocks.

Résumé : Les espèces de *Pandalus* présentent les caractéristiques suivantes qui compliquent l'application de modèles d'évaluation des stocks traditionnels reposant sur l'âge : (*i*) la difficulté de déterminer l'âge en l'absence de préservation des parties dures après la mue, (*ii*) le changement de sexe selon lequel, durant la maturation, les individus sont d'abord mâles pour ensuite se transformer en femelles, (*iii*) l'influence potentiellement forte des conditions ambiantes sur la dynamique de recrutement aux populations. Dans ce contexte, nous proposons un modèle d'évaluation saisonnier structuré par la taille pour l'évaluation des stocks de pandalidés hermaphrodites. Le cadre de modélisation intègre un sous-modèle pour les changements de longueur au moment de la transformation sexuelle et des fonctions pour incorporer des effets environnementaux sur la dynamique du recrutement. Le modèle peut être directement calé sur des données structurées selon la longueur, ce qui élimine le problème de conversion de la longueur en âge. Le modèle comprend un pas de temps saisonnier qui permet de tenir compte des variations saisonnières des processus biologiques et des habitudes de pêche. Il produit des paramètres d'évaluation des stocks comme des estimations de la mortalité par pêche et de la biomasse du stock, ainsi que l'abondance en fonction de la longueur selon le sexe. Le modèle est appliqué au stock exploité de crevettes *Pandalus borealis* dans le golfe du Maine pour en illustrer l'utilité. Le modèle proposé est souple et générique et peut être appliqué à de nombreux autres stocks exploités. [Traduit par la Rédaction]

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Introduction

Shrimp of the genus *Pandalus* are widely distributed in the North Atlantic and the North Pacific oceans. They are the main target of pandalid shrimp fisheries in the Northern Hemisphere (Garcia 2007). However, because of their unique life history characteristics, fisheries scientists and managers face great challenges in their stock assessment and management. Age-based assessment methods may not be suitable because direct determination of age is difficult for *Pandalus* stocks. Methods for aging using the eyestalk have been developed (Kilada et al. 2012), but further work is needed for practical application (NIPAG 2015). Approximated ages can be identified by examining the modes in length–frequency distributions in conjunction with some biological characteristics (Bergström 2000). However, this approach is uncertain because, frequently, modes overlap, and a single mode may actually contain more than one age group (Parsons et al. 1989). As a result, simple models that do not require age-disaggregated data are often used; these models fail to take full advantage of data available from fishery-dependent and fishery-independent sampling.

Sex change is a striking life history feature of most species of the genus *Pandalus*. They are hermaphroditic, maturing first as males and then transforming to females (Berkeley 1931; Bergström 2000). Because females are relatively large, they are the primary target of fisheries, and understanding sex change is considered critical for fisheries management (Fu et al. 2001). However, this biological process is seldom considered in stock assessment or fishery management even though sex stock structure can be greatly altered by fisheries. The factors that trigger sex change are not clearly understood, although many studies have investigated potential influences. Effects of temperature and stock size on length at sex change have been frequently reported in previous studies (Skúladóttir and Pétursson 1999; Hansen and Aschan 2000; Koeller et al. 2000; Wieland 2004). For example, a significant neg-

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ative correlation was found between the length at sex change and mean bottom temperature off West Greenland (Wieland 2004). According to sex allocation theory (Charnov 1982), pandalid length at sex change should decrease during periods of low female abundance to compensate for decreased reproductive capacity. Variability in length at sex change makes modeling this dynamic process important in assessment and management. A size-structured assessment model provides an ideal framework to facilitate modeling size-dependent life history processes.

Pandalus stocks are sensitive to changes in environmental conditions (Apollonio et al. 1986; Bergström 2000). Temperature plays an important role in regulating individual growth rates (Koeller et al. 2003; Daoud et al. 2010) and recruitment dynamics (Richards et al. 1996, 2012; Wieland and Siegstad 2012; Jónsdóttir et al. 2013). Growth varies both seasonally and interannually, with slower growth occurring in winter (Haynes and Wigley 1969; Pauly and Gaschutz 1978; Bergström 1992) and in populations in colder water (Shumway et al. 1985; Bergström 2000). In the Gulf of Maine (GOM), the stock–recruitment relationship of *Pandalus borealis* is influenced by temperature, with higher recruitment produced under cooler temperatures (Richards et al. 2012). In other *P. borealis* stocks, recruitment is related to temperature as well as predator biomass (Wieland and Siegstad 2012; Jónsdóttir et al. 2013). However, these important relationships are not explicitly considered in most assessment models applied to *P. borealis* (NEFSC 2014; NIPAG 2015).

Few analytical methods have been developed for shrimp stock assessments. Simple models that do not track cohorts, such as Catch–Survey–Analysis (Cadrin et al. 1999; Cadrin 2000), and surplus production models (Cadrin et al. 2004; NIPAG 2015) have been used for assessment. However, these models cannot take advantage of detailed demographic information, much less account for distinct biological and ecological characteristics such as sex change, environmentally driven recruitment, and seasonal or interannual growth variation. A new sex-specific assessment model for protandric pandalids was developed recently (Drouineau et al. 2012) that does include more detailed demographic information, but is not fully size-based or configured to account for environmental effects.

This paper presents a seasonal, size-structured assessment model for *Pandalus* that incorporates (*i*) a submodel for modeling changes in length at sex change and (*ii*) functions to incorporate environmental effects on recruitment dynamics. The model has a seasonal time step (number and duration of seasons can be defined), which makes it possible to account for seasonal variations in biological processes (e.g., growth and reproduction) and fishing patterns (e.g., a seasonal fishery). The model aims to improve pandalid stock assessment by increasing biological realism in simulating key life history and fishery processes. This paper includes two components: (*i*) description of the proposed size-structured stock assessment model and (*ii*) an application of the proposed model to the exploited GOM *P. borealis* stock. This study is part of a synthetic study that focuses on population dynamics modeling and assessment of *Pandalus* stocks.

Materials and methods

Integrated size-structured assessment model

The stock assessment model developed in this study is an integrated size-structured model that makes use of multiple data types. The model represents the population using a prespecified number of length-classes (k). It includes a population dynamics model that describes how the population changes over time and a series of observational models used to link the population dynamics model to the data observed in the survey and fishery. The basic dynamics equation can be expressed as

$$\mathbf{N}_m = \mathbf{G}_{m-1}\mathbf{V}_{m-1}\mathbf{N}_{m-1} + \hat{\mathbf{R}}_m \quad (1)$$

where $\mathbf{N}_m$ is a vector of length k containing the numbers-at-length at the beginning of season m, $\mathbf{G}_m$ is the growth transition matrix that describes the probabilities of shrimp growing from a length-class into other length-classes during season m, $\mathbf{V}_m$ is the survival rate from fishing and natural mortality, and $\hat{\mathbf{R}}_m$ is a vector of length k containing the recruitment to each length-class at the beginning of season m.

The model has a seasonal time step to properly account for seasonal biological processes and fishing patterns. The number and length of the seasons are flexible. For instance, three seasons can be defined with 2, 5, and 5 months within each season. To make the model more general and powerful, multiple modeling options are provided for each subcomponent. The following subsections describe the options and specifications of each subcomponent. Symbols are defined in Table 1.

Initial conditions

The numbers-at-length k at the start of the first year ($N_{\text{int},k}$) specifies the state of the population when the model starts and can be modeled in multiple ways defined below:

i. Treated as estimable parameters or fixed at some values (e.g., numbers-at-length in equilibrium).

ii. Calculated as the product of prespecified proportions-at-length k (Pia_k) and an estimable parameter of the total initial numbers (N_{int}):

$$N_{\text{int},k} = \text{Pia}_k N_{\text{int}} \quad (2)$$

iii. Calculated as the product of Pia_k and N_{int}, Pia_k is assumed to have a normal or lognormal distribution. In this case, mean (N_{int}) and standard deviation (Pia_k) of the normal–lognormal distribution are the parameters to be estimated.

iv. Calculated as the product of Pia_k and N_{int}, Pia_k is assumed to have a mixture normal distribution consisting of three normal distributions to account for multiple modes.

For options *iii* and *iv*, proportions-at-length are rescaled to sum to 1.

Recruitment

Recruitment in season m, year y, and length-class k ($R_{m,y,k}$) is assumed to enter into the population at the beginning of each time step and is modeled as the product of annual recruitment (R_y) and the proportion of the annual recruitment that recruits to each season (λ_m) and each length-class (λ_k):

$$R_{m,y,k} = R_y \lambda_m \lambda_k \quad (3)$$

The length-classes used to define recruitment can be prespecified, and the proportion of recruitment in each length-class can be either prespecified or estimated as parameters. λ_m also has to be prespecified. The annual recruitments can be specified in multiple ways described below:

i. Estimated as free parameters and modeled as

$$R_y = \bar{R}\, e^{\text{Rdev}_y - \frac{\sigma_R^2}{2}} \quad (4)$$

where Rdev_y is the recruitment deviation of year y from the expected recruitment ($\bar{R}$) and treated as bounded parameters, which means their sum is zero, so that they are centered on the expected recruitment; σ_R is the standard deviation for recruitment in log space.

ii. Related to spawning stock biomass (SSB) according to a stock–recruitment relationship (Beverton–Holt or Ricker):

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Table 1. Symbols and descriptions of parameters or variables used in the model.

Symbol	Description
k	Length
y	Year
m	Season
$\boldsymbol{N}$	Abundance
$\boldsymbol{G}$	Growth transition matrix
$\boldsymbol{V}$	Survival rate
$\hat{\boldsymbol{R}}$	Recruitment vector of length k
N_{int}	Initial total abundance
$\boldsymbol{Pia}$	Initial length composition vector of length k
R_y	Realized recruitment in year y
λ_m	Proportion of recruitment to each season
λ_k	Proportion of recruitment to each length-class
$\overline{R}$	Averaged recruitment for all the modeling years
Rdev_y	Deviation in recruitment on a log scale in year y
σ_R	Standard deviation for recruitment deviations in log space
SSB_y	Spawning stock biomass in year y
α	Parameter of stock–recruitment relationship
β	Parameter of stock–recruitment relationship
$E_{i,y}$	Environmental index i in year y
θ_i	Coefficient of environmental index i
$P_{y,k}$	Proportion of shrimp of a length-class k that change sex to female in year y
$L_{50\%,y}$	Parameter of sex change function
Inter_y	Parameter of sex change function
Bf_y	Females biomass in year y
$W_{y,k}$	Weight at length k in year y
p_{SSB}	Proportion of total mortality that occurred before spawning
Z	Total mortality
L_∞	Parameter of von Bertalanffy growth function
K	Parameter of von Bertalanffy growth function
σ_{L_∞}	Parameter of von Bertalanffy growth function
σ_K	Parameter of von Bertalanffy growth function
ρ_b	Parameter of von Bertalanffy growth function
low	Minimum length of a length bin
up	Maximum length of a length bin
F	Fishing mortality
S	Selectivity
M	Natural mortality
b	Time block
w_y	Weighting factor for natural mortality by year
w_k	Weighting factor for natural mortality by length
$C^{\text{pred}}_{f,m,y,k}$	Estimated catch at length k for fleet f in year y, season m
$\text{CPUE}^{\text{pred}}_{f,m,y}$	Estimated catch per unit effort for fleet f in year y, season m
$q_{f,m,y}$	Catchability for fleet f in year y, season m
$B^{\text{exploit}}_{f,m,y}$	Estimated biomass available to fishery f in year y, season m
$\text{CPUE}^{\text{obs}}_{f,m,y}$	Observed CPUE for fleet f in year y, season m
H	Power parameter
$I^{\text{pred}}_{\text{ind},y}$	Estimated abundance index for survey ind in year y
$I^{\text{obs}}_{\text{ind},y}$	Observed abundance index for survey ind in year y
indmonth	The month in which the survey takes place
n_b	The number of time blocks

$$R_y = \frac{\alpha \text{SSB}_y}{\beta + \text{SSB}_y} e^{\text{Rdev}_y} \quad (5)$$

or

$$R_y = \alpha \text{SSB}_y e^{-\beta \text{SSB}_y} e^{\text{Rdev}_y} \quad (6)$$

where α and β are shape parameters, and SSB_y is the spawning stock biomass of year y.

iii. Related to spawning stock biomass according to a modified stock–recruitment relationship (i.e., Beverton–Holt or Ricker) with the addition of parameters accounting for environmental effects (Hilborn and Walters 1992):

$$R_y = \frac{\alpha \text{SSB}_y}{\beta + \text{SSB}_y} e^{\sum_k \theta_i E_{i,y}} e^{\text{Rdev}_y} \quad (7)$$

$$R_y = \alpha \text{SSB}_y e^{-\beta \text{SSB}_y} e^{\sum_k \theta_i E_{i,y}} e^{\text{Rdev}_y} \quad (8)$$

where θ_i are parameters relating the multiple environmental time series (E_i) to the recruitment deviation. This explicitly assumes that environmental variables are measured without error and contribute to the portion of the overall recruitment deviation.

iv. Specified as options *i* or *ii* and assuming its deviations can be described by environmental variability. In this case, the environmental index, which could be a single index or a linear combination of multiple indices, is treated as if it were a survey of annual recruitment deviations (Schirripa et al. 2009). Because Rdev is assumed to have a normal distribution with mean zero, the environmental index is centered before fitting. Therefore, the likelihood of the deviations is added to the total likelihood and expressed as

$$L_{\text{Rdev}} = \sum_y \left(\frac{E_y - \text{Rdev}_y}{\sigma_y} \right)^2 \quad (9)$$

where E_y is the environmental index in year y, and σ_y is the standard deviation of the observation error of the environmental time series.

Sex change

Sex change is assumed to be length-dependent, and the proportion of shrimp of a given length-class that change sex to females in a given year ($P_{y,k}$) is modeled by a logistic function:

$$P_{y,k} = \frac{1}{1 + e^{\left[-\frac{2 \log(3)}{\text{Inter}_y} (L_k - L_{50\%,y}) \right]}} \quad (10)$$

where $L_{50\%,y}$ and Inter_y are two sets of parameters to be estimated. $L_{50\%,y}$ is the length at which 50% of shrimp change sex to females in year y, and Inter_y represents the interquartile range of the shrimp length distribution in year y. Sex composition data are used to fit this model.

Female biomass in year y (Bf_y) is calculated as

$$\text{Bf}_y = \sum_k N_{y,k} W_{y,k} P_{y,k} \quad (11)$$

where $W_{y,k}$ is the weight at length-class k for a given year y. SSB_y is calculated based on the population abundance-at-length in year y ($N_{y,k}$), weight at length in year y ($W_{y,k}$), proportion of females at length k in year y ($P_{y,k}$), and the proportion of the total mortality within the year prior to spawning (p_{SSB}) as

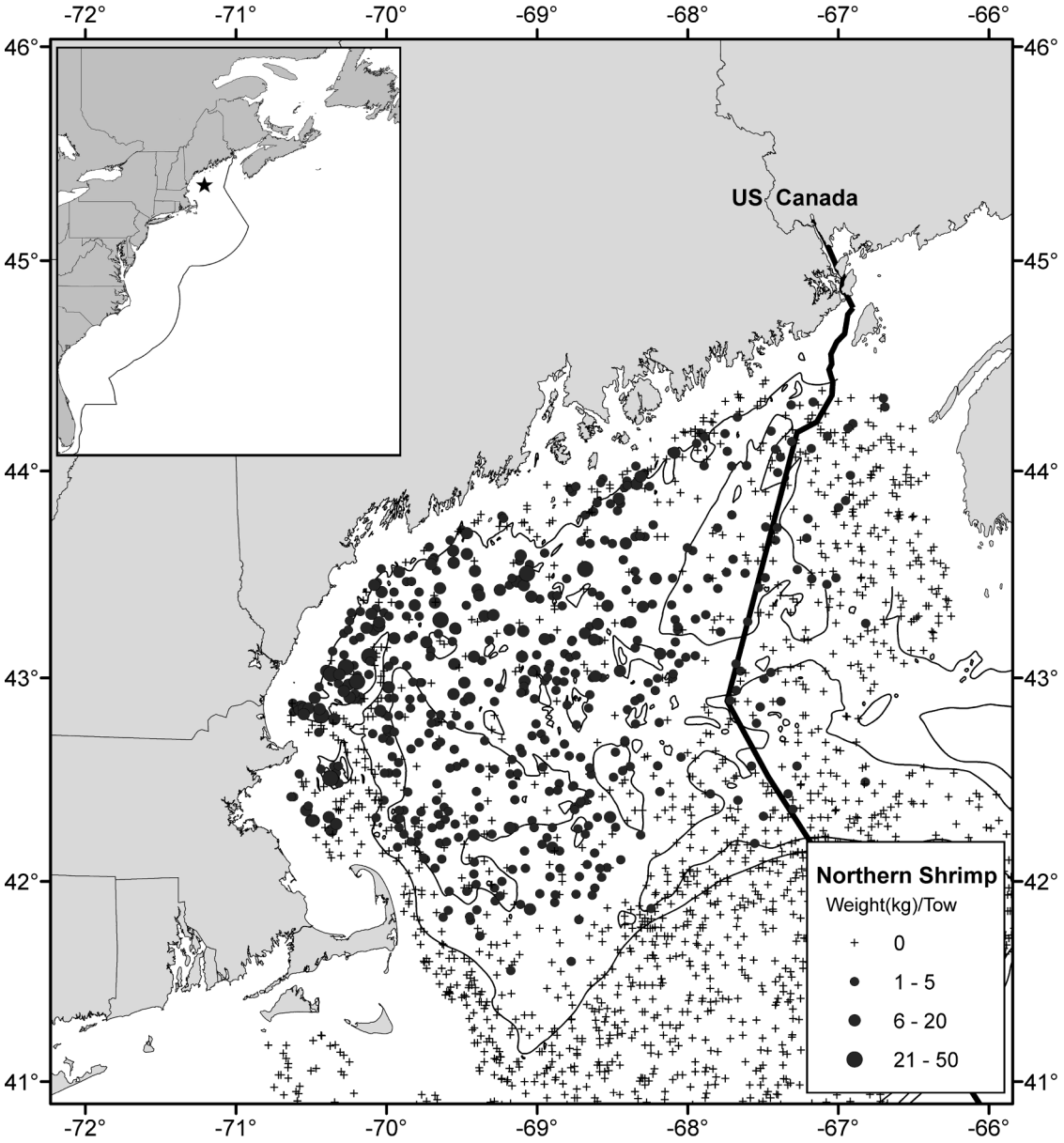
$$\text{SSB}_y = \sum_k N_{y,k} e^{-p_{\text{SSB}} Z_{y,k}} W_{y,k} P_{y,k} \quad (12)$$

Growth

The growth transition matrix which is required by the size-structured model can be either prespecified or estimated simultaneously with the other parameters in the model. The advantage of estimating it is that the resulting matrix will be consistent with other information included in the assessment. The growth matrix is modeled assuming that the expected growth increment follows the von Bertalanffy growth curve. The distribution of the growth

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Fig. 1. Map of *Pandalus borealis* stock distribution in the Gulf of Maine based on Northeast Fisheries Science Center (NEFSC) autumn trawl surveys, 1991–2008.



increment is assumed to be normal with mean $E(\Delta L_k)$ and variance $\mathrm{Var}(\Delta L_k)$, calculated as follows (Chen et al. 2003):

$$E(\Delta L_k) = (L_\infty - L_k)(1.0 - e^{-K}) \quad (13)$$

$$\mathrm{Var}(\Delta L_k) = \sigma^2_{L_\infty}(1 - e^{-K})^2 + (L_\infty - L_k)^2\sigma^2_K e^{-2K} + 2\rho_b\sigma_{L_\infty}\sigma_K(1 - e^{-K_b})(L_\infty - L_k)e^{-K} \quad (14)$$

L_∞, K, σ_{L_∞}, σ_K, and the correlation between L_∞ and K (ρ_b) are the parameters to be estimated. The probability of a shrimp growing from length-class k to length-class $k + 1$ ($Pp_{k\to k+1}$) is calculated as

$$Pp_{k\to k+1} = \int_{\mathrm{low}}^{\mathrm{up}} f[x|E(\Delta L_k), \mathrm{Var}(\Delta L_k)]dx \quad (15)$$

where "low" and "up" are the lower and upper ends of length-class $k + 1$, respectively.

The model with a seasonal time step allows growth to occur multiple times during the year. Thus, seasonal variation in growth can be captured, which increases biological realism. The model also allows interannual variation in growth by setting time blocks in which the growth is identical. The maximum number of time blocks is the number of time steps, meaning that a time-step-specific growth transition matrix could be specified. However, such a model configuration would require the estimation of more parameters.

Natural and fishing mortality

Fishing mortality is assumed to be separable and modeled as the product of a year effect (Fmult) and selectivity at length (S). Multiple fleets can be included in the model. The fishing mortality for a fleet f, year y, season m, and length bin k ($F_{f,m,y,k}$) is calculated as

$$F_{f,m,y,k} = \mathrm{Fmult}_{f,m,y}S_{f,m,k} \quad (16)$$

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Table 2. Summary of data input and model configurations for the Gulf of Maine northern shrimp stock assessment.

Item	Descriptor	Note
Years covered	1984–2013	All years with survey data
Seasons	2	Fishing season (December–May) and nonfishing season (June–November)
No. of sexes	2	Female and non-female
Lengths	10–35 mm	
Length bins	1 mm	
Weight at length	Exponential function	Ln(weight) = –7.4302 + 3.007(length) (from Haynes and Wigley 1969)
Initial condition	First-year length composition assumed in the model	Fall survey length composition in 1983
Recruitment lengths	10–12 mm	
Spawner–recruit relationship	No functional relationship	Recruitment deviations were tuned using an environment index; a likelihood function was used
Size at sex change	Proportion female at length	Data from ASMFC summer survey; incorporates a likelihood function to estimate the proportion of female
Growth	Growth transition matrix used in the model	Growth was assumed to be different by season; parameters were estimated
Commercial fleets	3	Mixed gear (1984–1999), trawl (2000–2013), trap (2000–2013)
Commercial selectivity at length	Mixed fleet (1984–1999)	Logistic
	Trawl fleet (2000–2013)	Logistic
	Trap fleet (2000–2013)	Logistic
Fishing mortality	Instantaneous rates	
Survey data	ASMFC summer	1984–2013
Survey selectivity at length	ASMFC summer	Logistic
Natural mortality	Time-varying	Constant over size
Likelihood weights	All one (1.0)	Used to weight each term in the negative log-likelihood

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The Fmult for a fleet f, year t, and season m is determined by two sets of parameters: $\text{Fmult}_{f,m,1}$, the parameter for the first year and each season for that fleet, and $\text{FDev}_{f,m,y}$, the deviation of the parameter from the value in the first year for that fleet. Both sets of parameters are estimated in log space:

$$\log(\text{Fmult}_{f,m,y}) = \log(\text{Fmult}_{f,m-1,y}) + \log(\text{FDev}_{f,m,y}) \quad (17)$$

For a given fleet, multiple time blocks can be specified to allow time dependence. Within each time block (b), there are four options for modeling selectivity for fleet f ($S_{f,b}$):

i. Estimate selectivity for each length bin
ii. Use logistic function (two parameters)
iii. Use double logistic function (four parameters)
iv. Use double normal function (Methot and Wetzel 2013)

The selectivity vector is rescaled to a maximum value of 1.0 for each defined selectivity time block.

Natural mortality (M) either can be assumed constant over length or predefined as time- or length-specific using weighting factors. The weighted M for shrimp of length-class k, in year y, season m ($M_{m,y,k}$) is calculated as

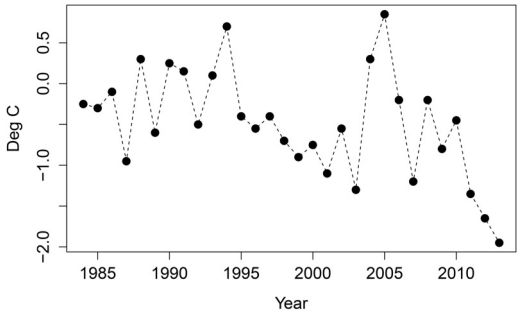
$$M_{m,y,k} = w_y w_k M_m \quad (18)$$

where w_y is a prespecified annual weighting factor, w_k is a prespecified length weighting factor, and M_m is seasonal natural mortality, which can be either prespecified or estimated.

Model predictions

Given population numbers and fishing mortality, expected catch in numbers of shrimp for each fleet, year, season, and length-class ($C^{\text{pred}}_{f,m,y,k}$) is calculated using the traditional Baranov equation:

Fig. 2. Mean bottom and surface temperature anomalies during NEFSC spring surveys, lagged 1 year. The additive inverse (negative) is shown because recruitment is negatively related to spring temperatures.



$$C^{\text{pred}}_{f,m,y,k} = \frac{F_{f,m,y,k}}{\sum_f F_{f,m,y,k} + M_{k,y,m}}[1 - e^{-(\sum_f F_{f,m,y,k} + M_{k,y,m})}]N_{m,y,k} \quad (19)$$

Catch per unit effort (CPUE) data can be included in the model, and the predicted CPUE in weight for fleet f, year y, and season m is calculated as

$$\text{CPUE}^{\text{pred}}_{f,m,y} = q_{f,m,y}\left(B^{\text{exploit}}_{f,m,y}\right)^H \quad (20)$$

where $q_{f,m,y}$ is the catchability for fleet f, year y, and season m, and H is the power parameter accounting for nonlinearity. $B^{\text{exploit}}_{f,m,y}$ is calculated as

$$B^{\text{exploit}}_{f,m,y} = \sum_k N_{m,y,k} \frac{1 - e^{-(\sum_f F_{f,m,y,k} + M_{m,y,k})}}{\sum_f F_{f,m,y,k} + M_{m,y,k}} S_{f,k} W_{y,k} \quad (21)$$

Table 3. Estimation option and unknown parameters to be estimated in the stock assessment (the number of years is 30).

Type of parameter	Parameter	Estimate option	No. of parameters
Initial numbers	First-year total abundance, N_{int}	*ii*	1
Recruitment	$Rdev_t$	*iv*	30
	λ_k		2
Sex change	$L_{50\%,t}$		30
	$Inter_y$		1
Growth	L_∞		2
	K		2
	σ_{L_∞}		2
	σ_K		2
	ρ_b		2
Commercial fishery	Selectivity, two parameters for logistic curve, three fleets	*ii*	6
	Fleet-specific fishing mortality		44
Survey	Selectivity, two parameters for logistic curve	*ii*	2
Total			126

Time blocks can be set up for fishery catchability to account for time dependence. The fishery catchability for time block b, fleet f, and season m ($q_{f,m,b}$) is calculated internally as

$$(22)\qquad q_{f,m,b} = \frac{1}{n_b}\sum_b \frac{\mathrm{CPUE}^{\mathrm{obs}}_{f,m,y}}{\left(B^{\mathrm{exploit}}_{f,m,y}\right)^H}$$

The survey abundance index ind in year y ($I^{\mathrm{pred}}_{\mathrm{ind},y}$) is calculated as

$$(23)\qquad I^{\mathrm{pred}}_{\mathrm{ind},y} = q_{\mathrm{ind}}\sum_k N_{y,k}[1 - \mathrm{e}^{-(\mathrm{indmonth}/12)Z_{y,k}}]S_{\mathrm{ind},y,k}$$

where indmonth refers to the end of month, and q_{ind} is the catchablity for suvey ind. Survey catchability for a given time block b ($q_{\mathrm{ind},b}$) is calculated internally as

$$(24)\qquad q_{\mathrm{ind},b} = \frac{1}{n_b}\sum_b \frac{I^{\mathrm{obs}}_{\mathrm{ind},y}}{\sum_k N_{y,k}[1 - \mathrm{e}^{-(\mathrm{indmonth}/12)Z_{y,k}}]S_{\mathrm{ind},y,k}}$$

Observational model and likelihood function

Observational models are developed to relate the model predictions with the observations made in the fishery and surveys. The model is fitted by maximizing the likelihood function that is derived from the observation models that describe the uncertainties around observed data. For catch and survey length composition, a multinomial distribution (eq. 25) or robust normal for proportions (eq. 26; Fournier et al. 1990) can be assumed. The robust normal likelihood limits the influence of high and low frequencies. Likelihood functions for normal (eq. 27), lognormal (eq. 28), and t distributions (eq. 29) are provided for total catch, CPUE, survey indices, and recruitment deviations. Likelihood functions on a log scale are listed as below:

$$(25)\qquad \ln(P) = \ln(\mathrm{ESS}!) - \sum_k (x_k!) + \mathrm{ESS}\sum_k p^{\mathrm{obs}}_k \ln\left(p^{\mathrm{pred}}_k\right)$$

$$(26)\qquad \ln(P) = \sum_k \left(-\ln\left\{\frac{1}{\mathrm{ESS}}\sqrt{2\pi\left[p^{\mathrm{pred}}_k\left(1 - p^{\mathrm{pred}}_k\right) + \frac{0.1}{\mathrm{SN}}\right]}\right\} + \ln\left\{\mathrm{e}^{-\frac{\left(p^{\mathrm{obs}}_k - p^{\mathrm{pred}}_k\right)^2}{2\left[p^{\mathrm{pred}}_k\left(1-p^{\mathrm{pred}}_k\right)+\frac{0.1}{\mathrm{SN}}\right]}} + 0.01\right\}\right)$$

$$(27)\qquad \ln(P) = -\ln\left(\sqrt{2\pi}\sigma^{\mathrm{obs}}_I\right) - \frac{\left(I^{\mathrm{obs}} - I^{\mathrm{pred}}\right)^2}{2\left(\sigma^{\mathrm{obs}}_I\right)^2}$$

$$(28)\qquad \ln(P) = -\ln\left(\sqrt{2\pi}\sigma^{\mathrm{obs}}_I\right) - \ln(I^{\mathrm{obs}}) - \frac{\left[\ln(I^{\mathrm{obs}}) - \ln(I^{\mathrm{pred}})\right]^2}{2\left(\sigma^{\mathrm{obs}}_I\right)^2}$$

$$(29)\qquad \ln(P) = -\ln\left(\frac{1.32934}{\sqrt{4\pi}}\right) - 2.5\ln\left\{\frac{\left[\ln(I^{\mathrm{obs}}) - \ln(I^{\mathrm{pred}})\right]^2}{2\left(\sigma^{\mathrm{obs}}_I\right)^2} + 1\right\}$$

Parameter estimation

A large number of parameters can be estimated. They are estimated by maximizing the log-likelihood that is a function of several components (e.g., total commercial catches, total abundance indices from survey, and length composition of commercial and survey catches). Weights can be given to each component to account for data quality. Priors can be specified for each parameter. Optimization is implemented using Automatic Differentiation Model Builder (ADMB; http://admb-foundation.org/), which minimizes the negative log-likelihood.

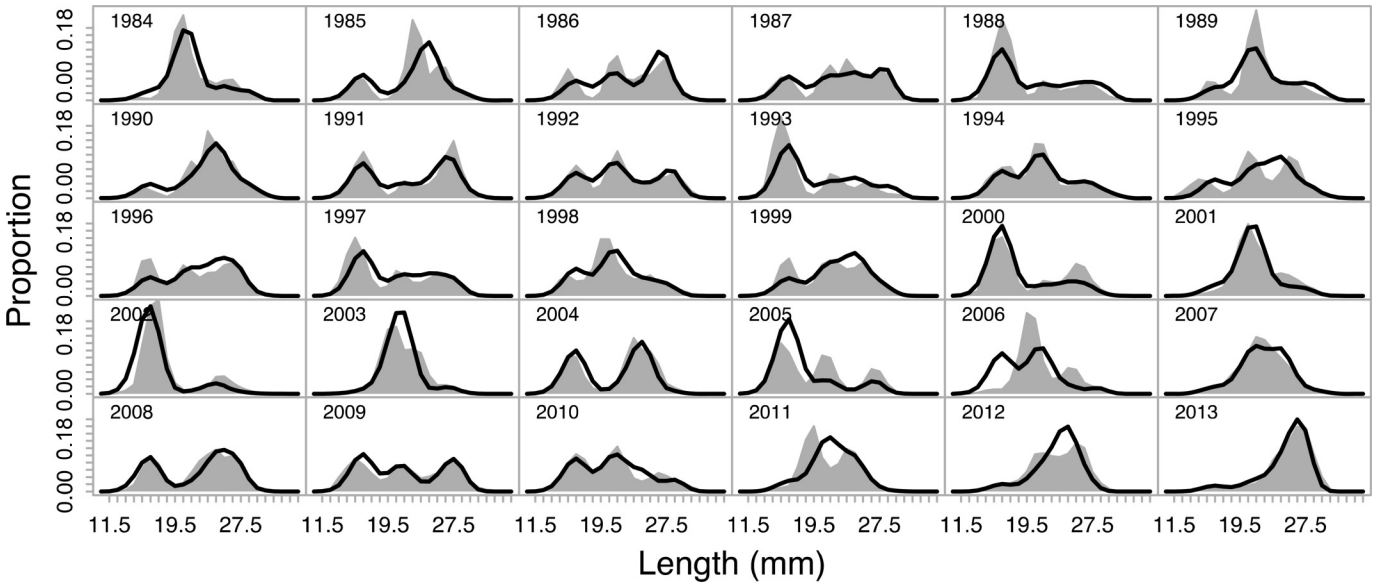
Case study: Gulf of Maine *P. borealis* stock

Description of the fishery and biology

The directed fishery for northern shrimp (*P. borealis*) in the GOM (Fig. 1) began in 1938 (Wigley 1973). Since then the fishery has been a dynamic one (Clark et al. 2000). The fishery is managed by total allowable catch and effort restrictions including a fishing season limited to months within a December to May time frame (a fishing year includes the previous December; e.g., the 2013 season includes December 2012 but not December 2013; ASMFC 2004; NEFSC 2014). The fishery peaks in midwinter when egg-bearing females move into inshore waters; therefore, the fishery mainly targets gravid individuals (Clark et al. 2000). The otter trawl is the primary gear employed in the fishery, but a minor trap fishery has also existed in midcoast Maine since the 1970s. The fishing season is shorter for trappers than for draggers because the trap fishery is dependent on the inshore availability of shrimp in a specific area. Commercial catches are sampled regularly to provide data on length and life history stage composition. Fishery-independent data have been collected since 1968 by the Northeast Fisheries Science Center (NEFSC) autumn bottom trawl survey and since 1984 by the Atlantic States Marine Fisheries Commission (ASMFC). The ASMFC survey is a stratified random survey conducted each summer and is directed specifically towards northern shrimp (Clark 1989).

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Fig. 3. Observed (gray polygons) and estimated (bold lines) annual length composition in fishery-independent surveys from 1984 to 2013 for the Atlantic States Marine Fisheries Commission (ASMFC) shrimp survey.



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Fig. 4. Observed (gray polygons) and estimated (bold lines) annual length composition of commercial catches from (*a*) 1985 to 1999 for the mixed gear fishery; (*b*) 2000 to 2013 for the trawl fishery, and (*c*) 2000 to 2013 for the trap fishery.

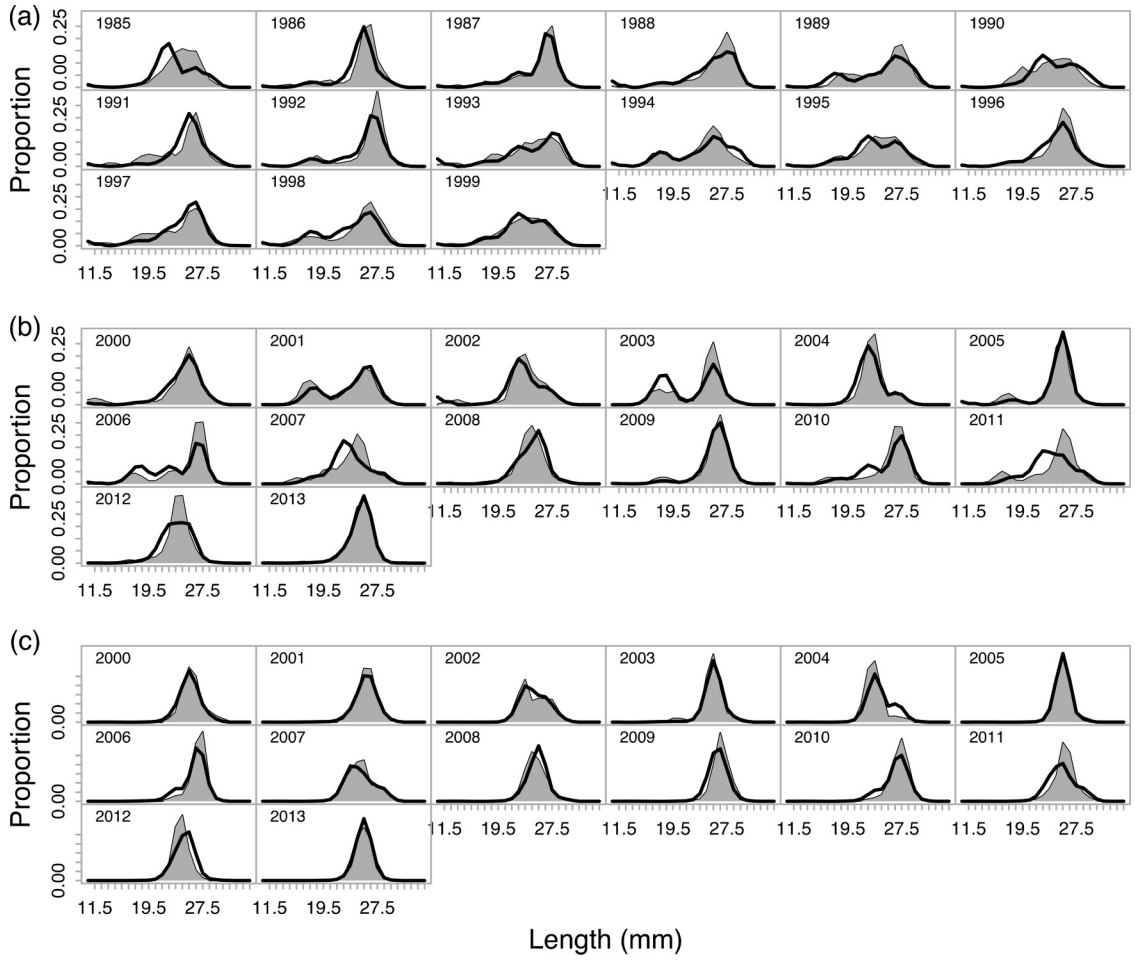


Fig. 5. Observed (black circles) and estimated (gray lines) commercial catches for (*a*) all fisheries; (*b*) mixed gear fishery; (*c*) trawl fishery; and (*d*) trap fishery.

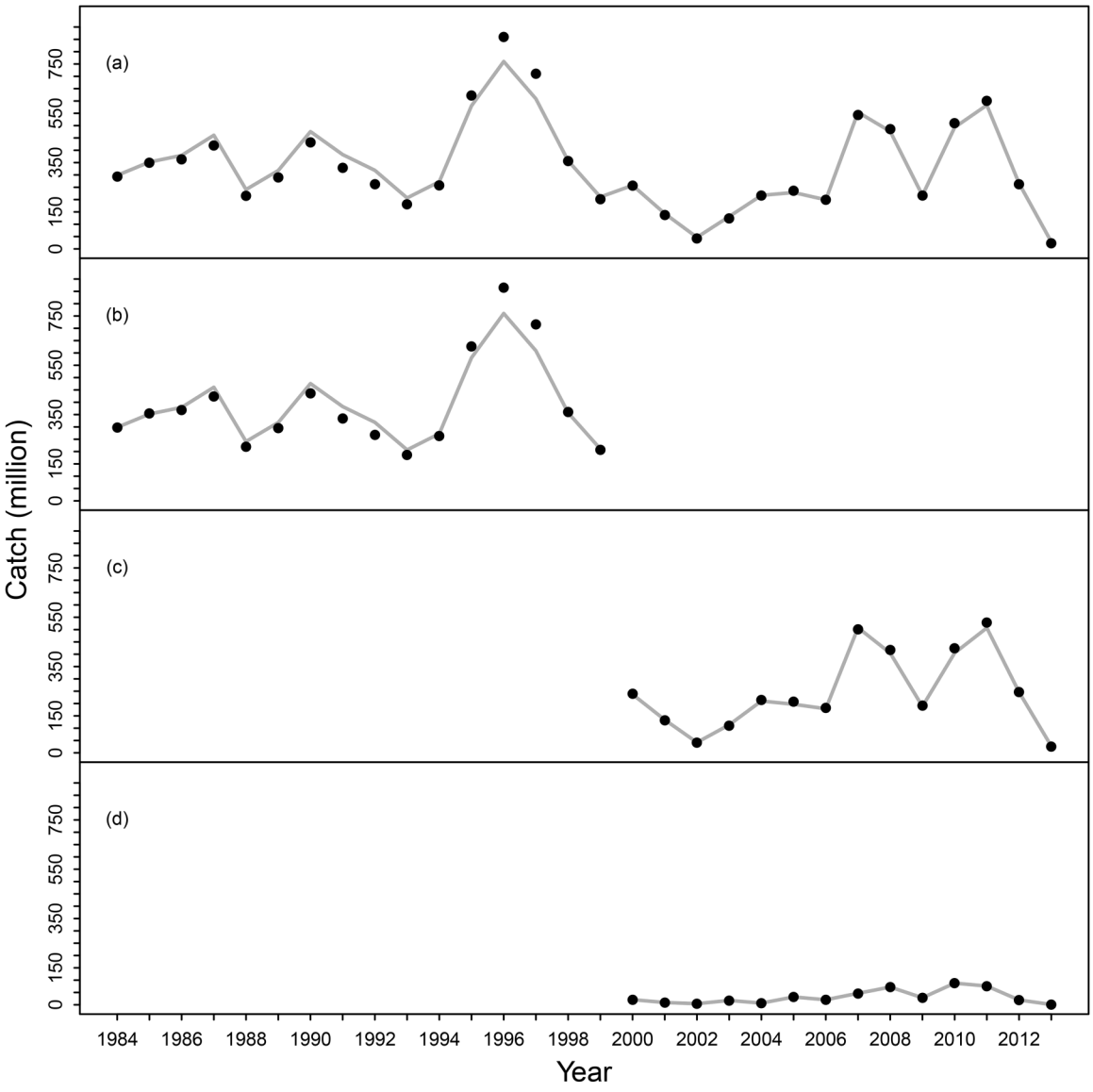
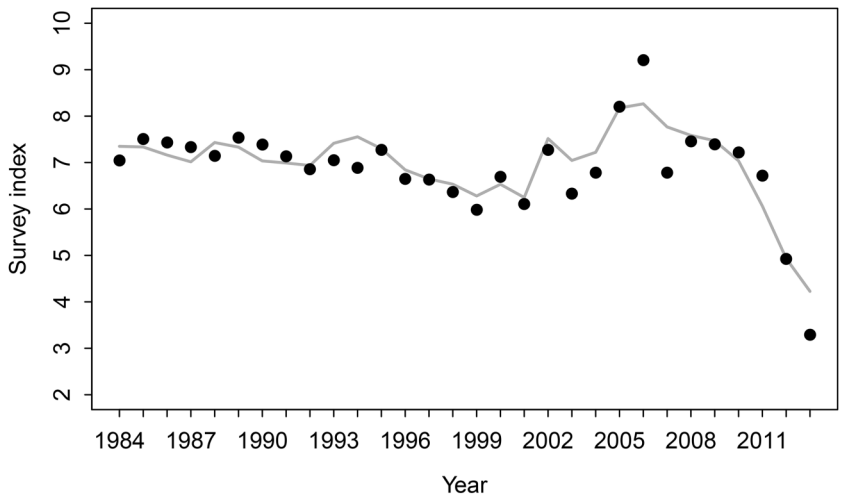
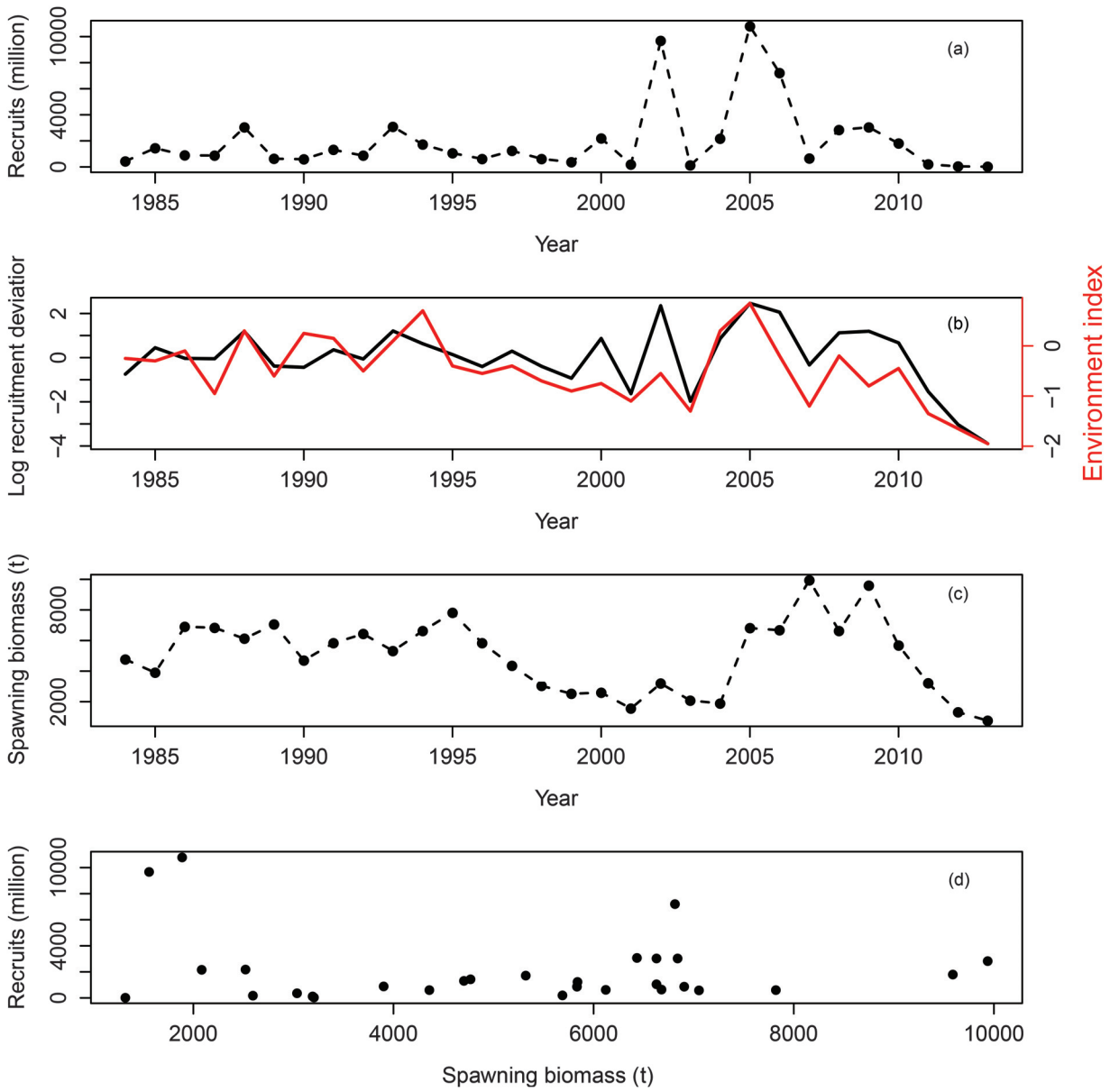


Fig. 6. Observed (black circles) and estimated (gray lines) survey abundances in logarithm space from ASMFC summer survey.



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Fig. 7. Estimated recruitment patterns: (*a*) estimated recruitment per year; (*b*) estimated log recruitment deviations and environment index; (*c*) estimated females per year; and (*d*) estimated stock–recruitment relationship: recruitments against previous year females. [Colour online.]



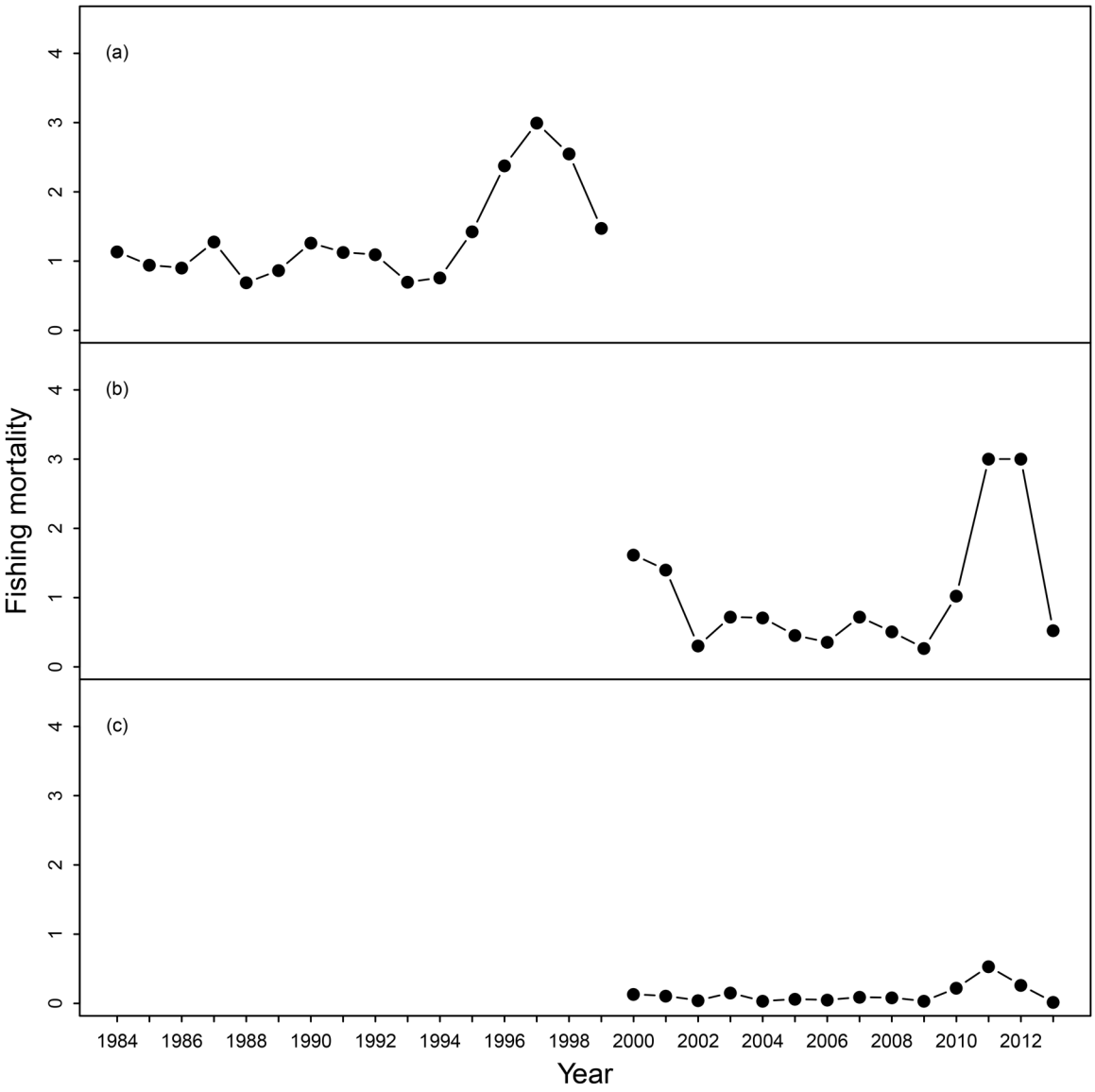
Northern shrimp in the GOM are at the southern extent of their range (Haynes and Wigley 1969) and are considered to be vulnerable to climate change effects, especially temperature. GOM northern shrimp usually mature as males at about 2.5 years of age and then transform to females at about 3 years of age (Clark et al. 2000), reproducing as females at ages 4 and 5, after which they are thought to die. Spawning takes place in offshore waters beginning in late July. Brooding females move inshore in late autumn and early winter and hatch their brood in inshore waters. Juveniles spend a year or more in coastal waters before migrating to offshore waters (Apollonio and Dunton 1969).

Input data and model configurations

Model input data and configurations for GOM shrimp are summarized in Table 2. The model was set to have a 6-month time step to account for seasonality in biology and the fishery. The first season extends from December through May (fishing season) and the second season from June through November (nonfishing season). The model was fitted over a 30-year data series (1984–2013). Commercial catch data were separated into three fleets to account for different fishery selectivities. Before 2000, the catch data cannot be separated by gear type (i.e., trawl and trap); therefore, fleet 1 in the model represents the mixed gear fishery from 1984 to 1999. Fleets 2 and 3 represent trawl and trap fisheries, respectively, during 2000–2013. Commercial catch-at-length data were available for the whole time period except 1984. Accordingly, three logistic functions were used to quantify fishing selectivity curves with the model parameters being estimated by the model. Data from the ASMFC summer shrimp survey (abundance indices and length frequency data, 1984–2013) were used in the assessment model, and fall survey data were used to inform the model initial length composition. Logistic functions were used to quantify the selectivity of the survey data with the parameters being estimated in the assessment. Proportion female at length from the summer survey was used to estimate annual size at sex transition. Carapace lengths were aggregated into 1 mm classes and ranged from 10.0 to 35.0 mm.

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Fig. 8. Estimated instantaneous fishing mortality (*F*) for (*a*) mixed gear fishery; (*b*) trawl fishery; and (*c*) trap fishery



Recruits are defined as shrimp of 10–12 mm carapace length (the first three length-classes) and are assumed to come into the population at the beginning of the first season (i.e., beginning of December). No functional relationship between recruitment and spawning stock was assumed. However, a time series of environmental indices was used to tune recruitment deviations (Fig. 2), which assumes that recruitment deviations are driven by environmental variables. The environmental index was the mean of 1-year lagged spring bottom and surface temperature anomalies, estimated from NEFSC spring surveys in offshore northern shrimp habitat (Richards et al. 2012). Recruitment is negatively related to spring temperatures in the GOM; therefore, we used the negative of the temperature anomalies to scale the recruitment deviations. Growth is assumed to differ by season but not interannually. The von Bertalanffy curve is assumed as a proxy for modeling the mean discontinuous growth experienced by shrimps. The growth parameters were estimated within the assessment model, which does not require estimating them externally using length-at-age data or identifying modes from length frequency data. Time-varying natural mortality (*M*) was used in the model. A baseline $M = 0.5 \cdot \text{year}^{-1}$ was scaled by an annual predation pressure index (Richards and Jacobson 2016), which incorporated the occurrence of pandalids in fish stomachs and predator biomass to derive an annual estimate of *M*. The estimated annual *M* was below average before the late 1990s, then increased and has fluctuated around higher levels since the late 1990s (Richards and Jacobson 2016). The initial length composition was fixed at the values observed in the fall survey in 1983 (model starts on 1 December 1983). A large number of parameters were estimated (Table 3). Parameters were grouped into two phases for estimation. Parameters of selectivity and sex change were estimated in phase 2, and the rest of the parameters were estimated in phase 1. Some specific parameter values were fixed. We set CV = 0.1 for catch of fleet 1 and CV = 0.05 for catch of fleets 2 and 3. The CV of the abundance indices of the summer survey was set to be 0.25.

Results

Fitting observed data

The model generally fit the length composition of the survey well (Fig. 3); however, for a few years with extremely narrow size modes (e.g., 1988, 1989, and 1993), the model could detect the modes but tended to predict lower values. The close fitting of length composition data suggests that the growth model is ade-

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quate. The length composition of landings was also well-fitted, especially when trawl and trap fleets could be modeled separately (Fig. 4). The narrow modes in the commercial length composition were well captured, demonstrating the adequacy of the fishery selectivity model. Temporal trends in total landings (in numbers) also fit well and so were the landings of each fishery (Fig. 5). The predicted landings were almost identical to the observed landings for all years except for 1996, in which landings were underestimated slightly. The trend in total survey abundance indices estimated by the model is consistent with the observations (Fig. 6). The survey detected a strong year effect for 2006; however, the model detected that year effect with much less magnitude. Overall, the model fit to the survey abundance index was not as good as the fit to landings, which is to be expected given the larger CV associated with the abundance index.

Population numbers and fishing mortality

Estimated recruitment varied after 2000 (Fig. 7*a*), with the highest recruitment occurring in 2005 and the lowest recruitment in 2013. The estimated recruitment in 2013 was only 0.2% of that in 2005. The recruitment deviations had temporal trends similar to the environmental index (Fig. 7*b*), indicating that the environmental driver might regulate recruitment dynamics and is consistent with other information in the model. SSB (i.e., female biomass) fluctuated greatly during the model time period especially after 1995 (Fig. 7*c*). SSB was at a low level in the late 1990s and early 2000s and then increased to high levels in 2005–2010. After 2010 it dropped dramatically to the lowest level in 2013 over the entire model period (727.7 t; less than 10% of its highest level in 2007). No obvious stock–recruitment relationship was observed (Fig. 7*d*). The strong year classes were not explained by a high level of female biomass, but were more likely due to environmental conditions favorable for larval survival, as the fitting of environmental indices was good.

The model provides fishing mortality estimates for each fishery in each fishing season, and three different exploitation rates were calculated (i.e., ratio of catch in number versus the total stock abundance, ratio of catch in weight versus total stock biomass, and ratio of female catch in weight versus female biomass). The estimated fishing mortality varied greatly over time, and traps resulted in much lower fishing mortality than trawls (Fig. 8). High levels of fishing mortality rates in the late 1990s and early 2010s explains the low female biomass during the same time period. The highest two exploitation rates occurred in 1997 and 2012 with 65.8% and 60.0% of females being removed in the fishery, respectively (Fig. 9). The recent decrease in the biomass of females resulted from high exploitation rates, continued low recruitments, and high natural mortality. Estimated fisheries selectivity for the three fishing fleets defined in the model showed differences, with traps more selective for larger shrimps. This is realistic because the trap fishery only targets female shrimp once they have migrated inshore. Compared with the estimated fisheries selectivities, the estimated survey selectivity had higher efficiency in catching small shrimps.

Sex change and growth

The model provides abundance estimates for non-females (males and transitionals) and females based on a time-varying sex change-at-length relationship. The model fits the sex proportion at length data well. The estimated length at which 50% of individuals become females ($L_{50\%}$) varied over time, with the highest and lowest $L_{50\%}$ occurring in 2008 and 2002, respectively (Fig. 10).

Apparent growth of a cohort under no fishing mortality estimated by the model shows large variation in length-at-age and a similar mean size for ages 4 and 5 (Fig. 11). This is consistent with the observation that size modes of large pandalids are often overlapped and hard to distinguish. Comparing the two extreme $L_{50\%}$ values with the mean size of the cohort for each year (Fig. 11)

Fig. 9. Estimated exploitation rates calculated as the ratio of predicted total catch in numbers and estimated total abundance (solid gray line); predicted total catch biomass and estimated total biomass (gray dotted line); and predicted total female catch in biomass and estimated total female biomass (black dotted line).

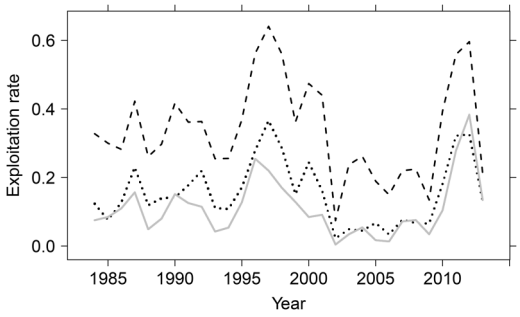


Fig. 10. Time series of model-estimated $L_{50\%}$ (size at sex change).

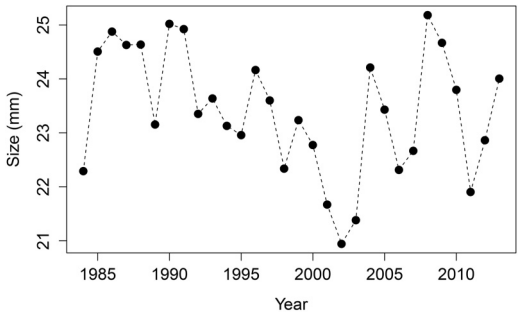
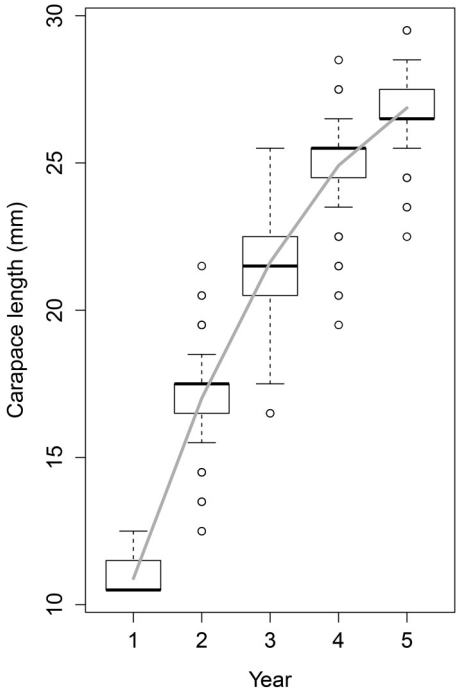
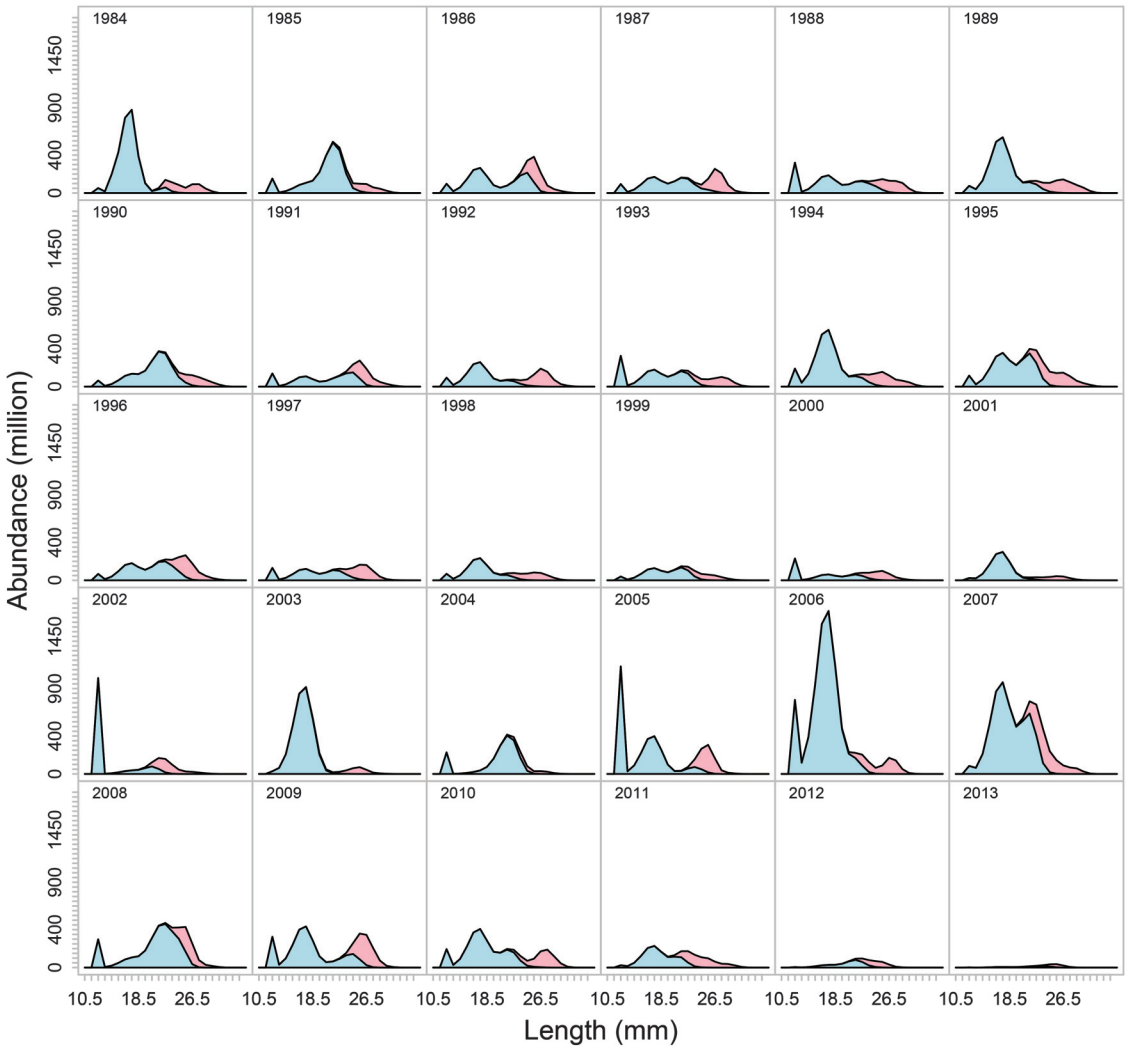


Fig. 11. Apparent length growth for a cohort estimated under no fishing mortality using the estimated growth transition matrices. Boxes represent the length distribution of that cohort in a particular year, and the gray line connects the means of each year.



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Fig. 12. Estimated sex-specific abundance-at-length. Pink polygons represent females and blue polygons represent non-females (males and transitionals). [Colour online.]



suggests that a significant proportion of males in 2008 may have become females at age 4 or even age 5, while those in 2002 may have changed sex at age 3 or even age 2. The sex-specific abundance-at-length estimates show that for most of the years, non-females tended to have two modes without overlapping with females (Fig. 12).

Discussion

This study presents an original size-structured stock assessment model developed for *Pandalus* stocks. This model addresses the specific life history processes of *Pandalus* species, which limit the use of traditional stock assessment models. The model can be directly fitted to length-structured data, overcoming the length-to-age conversion problem. It enables modeling sex change as a time-varying function of length rather than age, which is more appropriate for this species. Also, modeling sex change allows us to estimate population response (change in sex ratio at length) to changes in both natural and fishing mortality. Moreover, the model allows incorporation of environmental data to tune the recruitment estimates. The model proposed in this study is flexible and could be applied to many stocks, including other sequential hermaphrodites and gonochoristic species.

The flexibility of our model allows several alternative assumptions for most biological processes (e.g., recruitment can be estimated with or without relating to SSB or environment data; selectivity can be modeled as asymptotic or dome-shaped) and observation models. Time blocks can be set up for most of the parameters (e.g., growth parameters; selectivity parameters; catchability) to allow time dependence. The model time step is flexible, and the number of seasons and number of months within a season can be defined. Such flexibility in the model time step allows for variation in seasonality of fisheries and biology that may occur in different stocks. Therefore, the model presented in this study provides a generic stock assessment platform for *Pandalus* and other stocks.

Given the flexibility of the model, a single case study cannot demonstrate all the options built into the model. However, in a companion paper (Cao et al. 2017), the model was extensively tested through simulation before it was applied to the case study described here. The simulations were designed for evaluating performance of this model under various misspecification scenarios regarding recruitment dynamics, temporal growth pattern, fishing pattern, and natural mortality.

Modeling sex change within the assessment model is valuable for *Pandalus* management (Fu et al. 2001) because females are the primary target of commercial fisheries. Length at sex change tends to vary from year to year (Charnov and Anderson 1989; Fig. 10), and modeling that variation is also important for understanding population reproductive potential, especially because

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fecundity is related to female size (Bergström 2000). A size-structured model provides a good framework for modeling sex changes. Traditional assessment models for *Pandalus* stocks such as Catch–Survey–Analysis (Collie and Sissenwine 1983) do not integrate the sex change process; therefore, they do not provide sex structure information for management (Drouineau et al. 2012). Sex-specific abundance of the *Pandalus* stock in the GOM was known only from indices before applying our model. This model can estimate time-varying sex change-at-length (i.e., proportion of female-at-length). Thus, the exploitation rate, calculated as the ratio of harvested female biomass and total female biomass in this model, might be more realistic and accurate. Our model estimated a high interannual variability in sex change-at-length for *Pandalus* stock in the GOM (Fig. 10). Such high interannual variability is also seen in other *Pandalus* stocks (e.g., Gulf of St. Lawrence stock; Drouineau et al. 2012). The estimated interannual variability in sex change-at-length by the model could be utilized for further investigation (e.g., to test sex allocation theory; Charnov, 1982), which predicts that pandalid shrimp should decrease size at sex change during periods of low female abundance to compensate for decreased reproductive potential. We did not find a significant correlation between modeled female abundance and $L_{50\%}$ for our data, which may indicate more covariates are needed to explain the variation in $L_{50\%}$.

Environmental factors have long been hypothesized to play a major role in regulating annual fluctuations in exploited fish abundance, particularly through influences on the survival of early life stages (recruitment dynamics; Hjort 1914; Cushing 1982). Environmental variables rarely have been explicitly included in assessment models used for fishery management, although the approaches have been developed and tested (Maunder and Watters 2003; Schirripa et al. 2009). Given that recruitment of the GOM *Pandalus* stock has been demonstrated to be at least partially regulated by temperature (Richards et al. 2012, 2016), we believe that inclusion of environmental processes affecting recruitment into the assessment model would be beneficial. In our application, the estimated temporal trend in recruitment deviations was consistent with the lagged environmental index (Fig. 7*b*), suggesting that the environmental data were informative. However, simulation studies are needed to fully understand model performance under various scenarios (Chen et al. 2005) and test whether estimates of population parameters can be improved when environmental data are included in the model. It is important to understand the model performance if the environmental data are not informative, providing noise or even a misleading signal before we incorporate them into the model for practical applications.

Seasonal natural mortality is treated as a parameter in the model, which means that it can be fixed at a certain value or be estimated. Length-varying natural mortality rates can also be modeled. In our application, natural mortality was specified as time-varying but not length-varying. Although natural mortality is not likely to be constant over length (Lorenzen 1996), we had no precise information to allow us to employ length-varying natural mortality. Studies have been conducted to investigate the consequences of misspecification of natural mortality (Deroba and Schueller 2013; Johnson et al. 2015). Ignoring time-varying natural mortality leads to larger biases than ignoring age-specific natural mortality, suggesting that it is more important to account for time-varying natural mortality (Deroba and Schueller 2013). However, these studies were based on age-structured assessment models. The impacts of ignoring length-specific natural mortality for size-structured assessment model need to be evaluated.

A size-structured assessment model (Sex-Structured *Pandalus* Assessment Model, SPAM) was developed recently and applied to a *P. borealis* stock in Canada (Drouineau et al. 2012). SPAM can also describe the variability in size at sex change, but is not designed to account for the level of demographic detail or provide the flexibility of the model described here. In our model, the full population is divided into size classes, while SPAM divides the population into size-structured male and stage-structured female components. Therefore, SPAM cannot provide size composition information for females and assumes that the selectivity of females is equal to 1, which may be unrealistic. SPAM assumes constant growth and natural mortality over stages, while our model allows for growth to be time-varying and natural mortality to be time- and size-varying. Furthermore, our model has functions to include environmental data for recruitment estimation that are not accommodated in SPAM. To our knowledge, the model described here is the first size-structured assessment model for pandalid shrimp that incorporates environmental covariates.

The *P. borealis* stock in the GOM, the illustrative example in this paper, is considered collapsed, and the fishery has been closed since 2014 (NEFSC 2014). Assessment models historically applied to this stock (i.e., a biomass dynamics model and Catch–Survey–Analysis) were recently rejected as the basis for management (NEFSC 2014). Given the challenges of rebuilding a temperature-sensitive stock in the face of rapid environmental change (Mills et al. 2013), this newly developed model has the potential to provide improved assessment advice for effective fishery management.

Acknowledgements

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References

Apollonio, S., and Dunton, E.E., Jr. 1969. The northern shrimp, *Pandalus borealis*, in the Gulf of Maine. Maine Department of Sea and Shore Fisheries, Maine, USA.

Apollonio, S., Stevenson, D., and Dunton, E.E., Jr. 1986. Effects of temperature on the biology of the northern shrimp, *Pandalus borealis*, in the Gulf of Maine. NOAA/National Marine Fisheries Service, Maryland, USA.

ASMFC. 2004. Amendment 1 to the interstate fishery management plan for Northern shrimp. Atlantic States Marine Fisheries Commission, Virginia, USA.

Bergström, B. 1992. Growth, growth modelling and age determination of *Pandalus borealis*. Mar. Ecol. Prog. Ser. **83**(2–3): 167–183. doi:10.3354/meps083167.

Bergström, B.I. 2000. The biology of *Pandalus*. Adv. Mar. Biol. **38**: 55–245. doi:10.1016/S0065-2881(00)38003-8.

Berkeley, A.A. 1931. The post-embryonic development of the common pandalids of British Columbia. Contrib. Can. Biol. Fish. **6**(1): 79–163. doi:10.1139/f31-006.

Cadrin, S.X. 2000. Evaluating two assessment methods for Gulf of Maine Northern shrimp based on simulations. J. Northw. Atl. Fish. Sci. **27**: 119–132. doi:10.2960/J.v27.a11.

Cadrin, S.X., Clark, S.H., Schick, D.F., Armstrong, M.P., McCarron, D., and Smith, B. 1999. Application of catch-survey models to the Northern shrimp fishery in the Gulf of Maine. N. Am. J. Fish. Manage. **19**(2): 551–568. doi:10.1577/1548-8675(1999)019<0551:AOCSMT>2.0.CO;2.

Cadrin, S.X., Boutillier, J.A., and Idoine, J.S. 2004. A hierarchical approach to determining reference points for Pandalid shrimp. Can. J. Fish. Aquat. Sci. **61**(8): 1373–1391. doi:10.1139/f04-133.

Cao, J., Chen, Y., and Richards, R.A. 2017. Improving assessment of *Pandalus* stocks using a seasonal, size-structured assessment model with environmental variables. Part II: Model evaluation and simulation. Can J. Fish. Aquat. Sci. [In press.] doi:10.1139/cjfas-2016-0021.

Charnov, E.L. 1982. The theory of sex allocation. Princeton University Press, Princeton, New Jersey.

Charnov, E.L., and Anderson, P.J. 1989. Sex change and population fluctuations in pandalid shrimp. Am. Nat. **134**: 824–827.

Chen, Y., Hunter, M., Vadas, R., and Beal, B. 2003. Developing a growth-transition matrix for the stock assessment of the green sea urchin (*Strongylocentrotus droebachiensis*) off Maine. Fish. Bull. **101**(4): 737–744.

Chen, Y., Kanaiwa, M., and Wilson, C. 2005. Developing and evaluating a size-structured stock assessment model for the American lobster, *Homarus americanus*, fishery. N. Z. J. Mar. Freshwater Res. **39**(3): 645–660. doi:10.1080/00288330.2005.9517342.

For personal use only.

Clark, S.H. 1989. State-Federal northern shrimp survey. *In*: Proceedings of a workshop on bottom trawl surveys. Vol. 17. *Edited by* T.R. Azarovitz, J. McGurrin, and R. Seagraves. ASMFC, Virginia, USA. pp. 27–29.

Clark, S.H., Cadrin, S.X., Schick, D.F., Diodati, P.J., Armstrong, M.P., and McCarron, D. 2000. The Gulf of Maine northern shrimp (*Pandalus borealis*) fishery: a review of the record. J. Northw. Atl. Fish. Sci. **27**: 193–226. doi:10.2960/J.v27.a18.

Collie, J.S., and Sissenwine, M.P. 1983. Estimating population size from relative abundance data measured with error. Can. J. Fish. Aquat. Sci. **40**(11): 1871–1879. doi:10.1139/f83-217.

Cushing, D. 1982. Climate and fisheries. Academic Press, London. p. 373.

Daoud, D., Lambert, Y., Audet, C., and Chabot, D. 2010. Size and temperature-dependent variations in intermolt duration and size increment at molt of Northern shrimp, *Pandalus borealis*. Mar. Biol. **157**(12): 2655–2666. doi:10.1007/s00227-010-1526-1.

Deroba, J.J., and Schueller, A.M. 2013. Performance of stock assessments with misspecified age- and time-varying natural mortality. Fish. Res. **146**: 27–40. doi:10.1016/j.fishres.2013.03.015.

Drouineau, H., Savard, L., Desgagnés, M., and Duplisea, D. 2012. SPAM (Sex-Structured *Pandalus* Assessment Model): a stock assessment model for *Pandalus* stocks. Can. J. Fish. Aquat. Sci. **69**(4): 770–783. doi:10.1139/f2012-011.

Fournier, D.A., Sibert, J.R., Majkowski, J., and Hampton, J. 1990. MULTIFAN a likelihood-based method for estimating growth parameters and age composition from multiple length frequency data sets illustrated using data for southern bluefin tuna (*Thunnus maccoyii*). Can. J. Fish. Aquat. Sci. **47**(2): 301–317. doi:10.1139/f90-032.

Fu, C., Quinn, T.J., II, and Shirley, T.C. 2001. The role of sex change, growth and mortality in *Pandalus* population dynamics and management. ICES J. Mar. Sci. **58**(3): 607–621. doi:10.1006/jmsc.2001.1050.

Garcia, E.G. 2007. The northern shrimp (*Pandalus borealis*) offshore fishery in the Northeast Atlantic. Adv. Mar. Biol. **52**: 147–266. doi:10.1016/S0065-2881(06)52002-4. PMID:17298891.

Hansen, H.Ø., and Aschan, M. 2000. Growth, size- and age-at-maturity of shrimp, *Pandalus borealis*, at Svalbard related to environmental parameters. J. Northw. Atl. Fish. Sci. **27**: 83–91. doi:10.2960/J.v27.a8.

Haynes, E.B., and Wigley, R.L. 1969. Biology of the northern shrimp, *Pandalus borealis*, in the Gulf of Maine. Trans. Am. Fish. Soc. **98**(1): 60–76. doi:10.1577/1548-8659(1969)98[60:BOTNSP]2.0.CO;2.

Hilborn, R., and Walters, C.J (*Editors*). 1992. Quantitative fisheries stock assessment: choice, dynamics and uncertainty. Springer, New York. doi:10.1007/978-1-4615-3598-0.

Hjort, J. 1914. Fluctuations in the great fisheries of northern Europe viewed in the light of biological research. Rapp. P.-v. Réun. Cons. Int. Explor. Mer, **20**: 1–228.

Johnson, K.F., Monnahan, C.C., McGilliard, C.R., Vert-pre, K.A., Anderson, S.C., Cunningham, C.J. , et al. 2015. Time-varying natural mortality in fisheries stock assessment models: identifying a default approach. ICES J. Mar. Sci. **72**(1): 137–150. doi:10.1093/icesjms/fsu055.

Jónsdóttir, I.G., Magnússon, Á., and Skúladóttir, U. 2013. Influence of increased cod abundance and temperature on recruitment of northern shrimp (*Pandalus borealis*). Mar. Biol. **160**(5): 1203–1211. doi:10.1007/s00227-013-2172-1.

Kilada, R., Sainte-Marie, B., Rochette, R., Davis, N., Vanier, C., and Campana, S. 2012. Direct determination of age in shrimps, crabs, and lobsters. Can. J. Fish. Aquat. Sci. **69**(11): 1728–1733. doi:10.1139/cjfas-2012-0254.

Koeller, P., Mohn, R., and Etter, M. 2000. Density dependant sex change in Northern shrimp, *Pandalus borealis*, on the Scotian Shelf. J. Northw. Atl. Fish. Sci. **27**: 107–118. doi:10.2960/J.v27.a10.

Koeller, P.A., Covey, M., and King, M. 2003. Is size at sex transition an indicator of growth or abundance in Pandalid shrimp? Fish. Res. **65**: 217–230. doi:10.1016/j.fishres.2003.09.016.

Lorenzen, K. 1996. The relationship between body weight and natural mortality in juvenile and adult fish: a comparison of natural ecosystems and aquaculture. J. Fish Biol. **49**(4): 627–642. doi:10.1111/j.1095-8649.1996.tb00060.x.

Maunder, M.N., and Watters, G.M. 2003. A general framework for integrating environmental time series into stock assessment models: model description, simulation testing, and example. Fish. Bull. **101**(1): 89–99.

Methot, R.D., Jr., and Wetzel, C.R. 2013. Stock synthesis: A biological and statistical framework for fish stock assessment and fishery management. Fish. Res. **142**: 86–99. doi:10.1016/j.fishres.2012.10.012.

Mills, K.E., Pershing, A.J., Brown, C.J., Chen, Y., Chiang, F.-S., Holland, D.S., et al. 2013. Fisheries management in a changing climate: Lessons from the 2012 ocean heat wave in the Northwest Atlantic. Oceanography, **26**: 191–195. doi:10.5670/oceanog.2013.27.

NEFSC. 2014. 58th Northeast regional Stock Assessment Workshop (58th SAW) assessment report [online]. Northeast Fisheries Science Center, National Marine Fisheries Service, Woods Hole, Massachusetts. Available from http://nefsc.noaa.gov/publications/ [accessed November 2015].

NIPAG. 2015. Report of the NAFO/ICES *Pandalus* Assessment Group (NIPAG) Meeting, 9–16 September 2015, St. John's, Newfoundland, Canada. NIPAG, Newfoundland.

Parsons, D.G., Mercer, V.L., and Veitch, P.J. 1989. Comparison of the growth of Northern shrimp (*Pandalus borealis*) from four regions of the Northwest Atlantic. J. Northw. Atl. Fish. Sci. **9**: 123–131.

Pauly, D., and Gaschutz, G. 1978. A simple method of fitting oscillating length growth data, with a program for pocket calculators. International Council for the Exploration of the Sea, Denmark.

Richards, R.A., O'Reilly, J.E., and Hyde, K.J.W. 2016. Use of satellite data to identify critical periods for early life survival of northern shrimp in the Gulf of Maine. Fisheries Oceanography, **25**: 306–319. doi:10.1111/fog.12153.

Richards, R.A., and Jacobson, L.D. 2016. A simple predation pressure index for modeling changes in natural mortality: Application to Gulf of Maine northern shrimp stock assessment. Fish. Res. **179**: 224–236. doi:10.1016/j.fishres.2016.03.003.

Richards, R.A., Fogarty, M.J., Clark, S., Schick, D.S., Diodati, P., and O'Gorman, B. 1996. Relative influence of reproductive capacity and temperature on recruitment of *Pandalus borealis* in the Gulf of Maine. International Council for the Exploration of the Sea, Denmark.

Richards, R.A., Fogarty, M.J., Mountain, D.G., and Taylor, M.H. 2012. Climate change and northern shrimp recruitment variability in the Gulf of Maine. Mar. Ecol. Prog. Ser. **464**: 167–178. doi:10.3354/meps09869.

Schirripa, M.J., Goodyear, C.P., and Methot, R.M. 2009. Testing different methods of incorporating climate data into the assessment of US West Coast sablefish. ICES J. Mar. Sci. **66**(7): 1605–1613. doi:10.1093/icesjms/fsp043.

Shumway, S.E., Perkins, H.C., Schick, D.F., and Stickney, A.P. 1985. Synopsis of biological data on the pink shrimp, *Pandalus borealis* Krøyer, 1838. National Oceanic and Atmospheric Administration, Washington, DC.

Skúladóttir, U., and Pétursson, G. 1999. Defining populations of northern shrimp, *Pandalus borealis* (Krøyer 1938), in Icelandic waters using maximum length and maturity ogive of females. Rit Fiskid. **16**: 247–262.

Wieland, K. 2004. Length at sex transition in northern shrimp (*Pandalus borealis*) off West Greenland in relation to changes in temperature and stock size. Fish. Res. **69**(1): 49–56. doi:10.1016/j. fishres.2004.04.003.

Wieland, K., and Siegstad, H. 2012. Environmental factors affecting recruitment of northern shrimp (*Pandalus borealis*) in West Greenland waters. Mar. Ecol. Prog. Ser. **469**: 297–306. doi:10.3354/meps09794.

Wigley, R.L. 1973. Fishery for northern shrimp, *Pandalus borealis*, in the Gulf of Maine. Mar. Fish. Rev. **35**(3–4): 9–14.

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